

Why $-\log_2 P(X = i)$ is the Measure for Binary Representation of Messages

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Abstract

In information theory, the quantity $I(x) = -\log_2 P(X = x)$ represents the self-information of an event. This paper presents a rigorous mathematical proof establishing that $-\log_2 P(X = i)$ corresponds strictly to the theoretical minimum number of binary (yes/no) questions required to uniquely identify or represent a message generated with probability $P(X = i)$. We explore both the exact dyadic probability setting and the generalized asymptotic regime via Shannon's Source Coding Theorem.

1 Introduction and Setup

Let $\mathcal{X} = \{x_1, x_2, \dots, x_M\}$ be a finite alphabet of possible messages generated by a discrete random variable X , governed by a probability mass function $P : \mathcal{X} \rightarrow [0, 1]$, where $P(X = x_i) = p_i > 0$ and $\sum_{i=1}^M p_i = 1$.

We model the process of identifying a realized message $x_i \in \mathcal{X}$ through a sequence of binary questions (i.e., questions yielding an output in $\{0, 1\}$ or $\{\text{Yes}, \text{No}\}$).

Definition 1 (Binary Decision Tree and Code Length). *A binary query strategy over $\mathcal{X}$ is isomorphic to a prefix-free binary tree T , where:*

- (i) *Each internal node represents a binary question partitioning the remaining candidate space.*
- (ii) *Each leaf node corresponds uniquely to a message $x_i \in \mathcal{X}$.*
- (iii) *The depth of the leaf node x_i , denoted by $\ell_i \in \mathbb{N}^+$, corresponds to the exact number of binary questions required to uniquely determine x_i .*

2 Formal Proof

2.1 Axiomatic Characterization via Information Properties

Before constructing the search tree proof, we prove that any information measure $I(p)$ mapping a probability $p \in (0, 1]$ to the required number of binary questions must take the logarithmic form.

Theorem 1 (Uniqueness of the Logarithmic Measure). *Let $I(p)$ be a function representing the number of binary decisions required to specify an event occurring with probability p . If $I(p)$ satisfies:*

- A1. **Monotonicity:** $I(p_1) > I(p_2)$ for $p_1 < p_2$ (less probable events require more questions).
- A2. **Additivity for Independent Events:** $I(p_1 \cdot p_2) = I(p_1) + I(p_2)$.
- A3. **Normalization:** $I(\frac{1}{2}) = 1$ (a fair binary choice requires exactly 1 question).

Then $I(p) = -\log_2(p)$ for all $p \in (0, 1]$.

Proof. Let $p, q \in (0, 1]$. By Additivity (A2), $I(p \cdot q) = I(p) + I(q)$. Applying this inductively for any $m, n \in \mathbb{N}^+$ and $p = 1/2$:

$$I\left(\left(\frac{1}{2}\right)^m\right) = m \cdot I\left(\frac{1}{2}\right) = m$$

Now consider $p = (1/2)^{m/n}$. Taking n times the function:

$$n \cdot I\left(\left(\frac{1}{2}\right)^{m/n}\right) = I\left(\left(\frac{1}{2}\right)^m\right) = m \implies I\left(\left(\frac{1}{2}\right)^{m/n}\right) = \frac{m}{n}$$

By monotonicity (A1) and dense continuity of rational numbers in $\mathbb{R}^+$, for any general probability $p \in (0, 1]$:

$$I(p) = -\log_2(p)$$

□

2.2 Case 1: Dyadic Probabilities (Exact Integer Question Bounds)

Theorem 2 (Binary Decision Tree Equivalence for Dyadic Distributions). *Suppose $P(X = i) = 2^{-k_i}$ for some integer $k_i \in \mathbb{N}^+$ for all $i \in \{1, \dots, M\}$. Then the optimal number of binary questions ℓ_i to identify message x_i is precisely:*

$$\ell_i = -\log_2 P(X = i)$$

Proof. By Kraft's Inequality, a binary prefix code (or valid binary query tree) with word lengths $\ell_1, \ell_2, \dots, \ell_M$ exists if and only if:

$$\sum_{i=1}^M 2^{-\ell_i} \leq 1 \tag{1}$$

To minimize the expected number of questions $L = \sum_{i=1}^M p_i \ell_i$ subject to (1), we formulate the continuous optimization problem using Lagrange multipliers, relaxing ℓ_i to $\mathbb{R}^+$:

$$\mathcal{J}(\ell_1, \dots, \ell_M, \lambda) = \sum_{i=1}^M p_i \ell_i + \lambda \left(\sum_{i=1}^M 2^{-\ell_i} - 1 \right)$$

Taking partial derivatives with respect to ℓ_i :

$$\frac{\partial \mathcal{J}}{\partial \ell_i} = p_i + \lambda 2^{-\ell_i} \ln(2) = 0 \implies 2^{-\ell_i} = \frac{p_i}{-\lambda \ln 2}$$

Summing both sides over i :

$$\sum_{i=1}^M 2^{-\ell_i} = \frac{\sum_{i=1}^M p_i}{-\lambda \ln 2} = \frac{1}{-\lambda \ln 2}$$

For optimal tree utilization, the Kraft sum achieves equality $\sum 2^{-\ell_i} = 1$, yielding $-\lambda \ln 2 = 1$. Substituting back:

$$2^{-\ell_i} = p_i \implies \ell_i = -\log_2(p_i) = -\log_2 P(X = i)$$

Since $P(X = i) = 2^{-k_i}$ is dyadic, $\ell_i = k_i \in \mathbb{N}^+$, meaning this lower bound is achievable with an exact integer number of questions. □

2.3 Case 2: General Probabilities and Asymptotic Bounds

When $P(X = i)$ is not a power of $1/2$, the theoretical number of questions $-\log_2 P(X = i)$ is non-integer.

Theorem 3 (Shannon's Source Coding Theorem Bounds). *For any probability distribution $P(X)$, the optimal prefix-free code length ℓ_i^* satisfies:*

$$-\log_2 P(X = i) \leq \ell_i^* < -\log_2 P(X = i) + 1$$

Furthermore, when querying sequences of n independent identically distributed (i.i.d.) messages $X^n = (X_1, X_2, \dots, X_n)$, the average number of binary questions per message q_n satisfies:

$$\lim_{n \rightarrow \infty} q_n = \mathbb{E}[-\log_2 P(X)] = H(X)$$

Proof. By choosing $\ell_i = \lceil -\log_2 P(X = i) \rceil$, we ensure:

$$\sum_{i=1}^M 2^{-\ell_i} = \sum_{i=1}^M 2^{-\lceil -\log_2 p_i \rceil} \leq \sum_{i=1}^M 2^{\log_2 p_i} = \sum_{i=1}^M p_i = 1$$

Thus, Kraft's inequality holds and a valid binary decision tree exists. The exact query length for a single instance lies strictly within a 1-bit overhead window of $-\log_2 P(X = i)$.

For a sequence of length n , $P(X^n = (x_{i_1}, \dots, x_{i_n})) = \prod_{j=1}^n P(X = x_{i_j})$. The optimal code length L_n for the sequence satisfies:

$$-\log_2 P(X^n) \leq L_n < -\log_2 P(X^n) + 1$$

Dividing by n :

$$-\frac{1}{n} \sum_{j=1}^n \log_2 P(X = x_{i_j}) \leq \frac{L_n}{n} < -\frac{1}{n} \sum_{j=1}^n \log_2 P(X = x_{i_j}) + \frac{1}{n}$$

Taking expectations as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[L_n]}{n} = \mathbb{E}[-\log_2 P(X)]$$

This proves that $-\log_2 P(X = i)$ is the exact measure of binary question complexity per symbol. $\square$

Conclusion

We have proven that:

1. Under axiomatic information assumptions, $I(p) = -\log_2(p)$ is the unique additive and monotonic decision function.
2. For dyadic distributions, $-\log_2 P(X = i)$ gives the exact integer number of binary questions needed.
3. For arbitrary distributions, $-\log_2 P(X = i)$ represents the absolute lower bound on binary questions, achievable asymptotically per message without redundancy.